Math 9 Fall 2022 — Lecture #9 Worksheet
Corresponding Outcomes: GOMa3, GOMa4, GOMa5

This is an in-class worksheet. You do not need to turn in anything. The goal of this worksheet is to write a function `bisectnew(f,x1,x2)` that works the same as the bisect method function from our video lectures, but with the following differences. (Hint: If the original code for the bisect method doesn't make sense, definitely try to understand that first!)

1. The function bisectnew should work as long as $f(x1)$ and $f(x2)$ have different signs. (In other words, if $f(x1)$ is positive and $f(x2)$ is negative, the function should work, and if $f(x1)$ is negative and $f(x2)$ is positive, the function should also work.) If you find yourself copying and pasting a block of code, that is not a reasonable approach (because then, if you change the code in one section, you also have to remember to change it in the other section).

2. Use the 'error' command in Matlab to display an error if $f(x1)$ and $f(x2)$ have the same sign. The syntax to use is `error(msg)`, where msg is a string with the error message you want to display.

3. Instead of having an error threshold as input to bisectnew, use `1000·max(eps(x1),eps(x2))` as the error threshold.

4. Think about how you would answer the following:

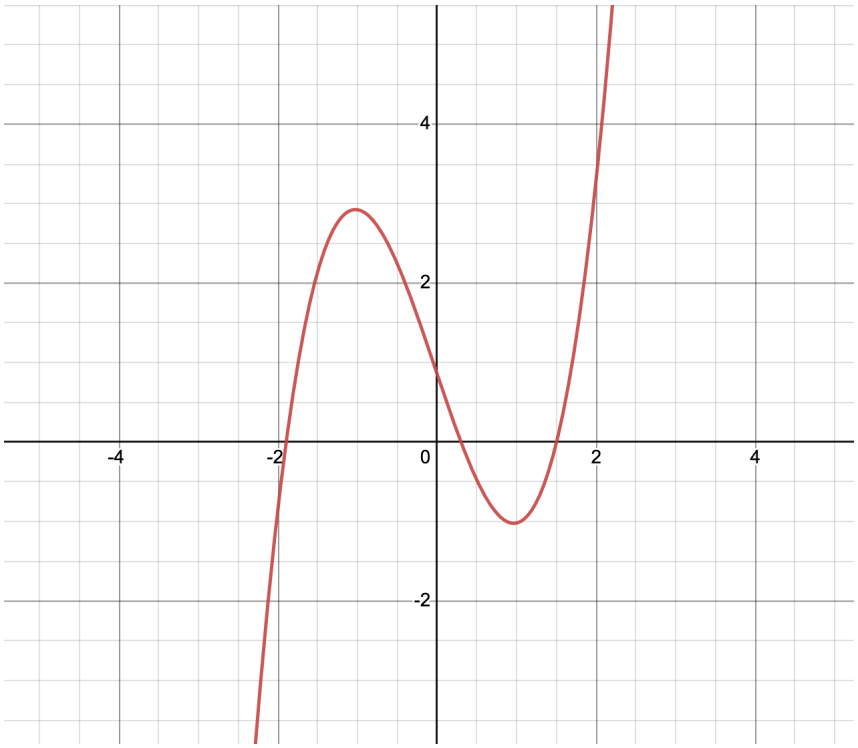
   (a) Setting the error threshold to be `min(eps(x1),eps(x2))` would not make sense; explain. (This is a good point to understand, but it also probably the least important question on this worksheet.)

   (b) Describe why the bisect method does not work if you apply it to the function $\tan(x)$ with the inputs $x = 1$ and $x = 2$.

   (c) If we apply the bisect method to the function $y = f(x)$ graphed below for the inputs $x = -5$ and $x = 5$, what zero is approximated? (The three zeros are approximately $x = -1.9, .3, 1.5$.)

   (d) If we apply the bisect method to the function $y = f(x)$ graphed below for the inputs $x = -11$ and $x = 5$, what zero is approximated? (The three zeros are approximately $x = -1.9, .3, 1.5$.)

   (e) If we apply the bisect method to the function $y = f(x)$ graphed below for the inputs $x = -1$ and $x = 5$, what zero is approximated? (Trick question.)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